Norges miljø- og biovitenskapelige universitet

BUS328 Final Exam 2023

School of Economics and Business

An investment analysis of Norwegian Solar Park

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Date: 11.05.2023

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Executive Summary

The paper provides a potential investment opportunity from the solar power company on a large scale industrial Solar Park located in the Western part of Norway. The plant expects to produce 130 MWh of energy utilizing sunlight directly. The project will be funded 20% by the venture capitalist firm in an exchange for an equity stake in the solar power company, and this gives potential rights for the capitalist firm to engage in the planning and the development of mitigation measures.

The report will cover the key sustainability issues regarding investment as well as possible mitigation measures. The Solar Park project promises to bring many opportunities such as the production of green and renewable energy, and the possibility of increased employment in the area. However, concerns over the environmental and social aspects require the emphasis on stakeholder dialogue and mitigation measures. One such adaptive method is to integrate biodiversity and promote friendly-pollinator environment in the park through sheep grazing.

The net present value analysis shows that the valuation is mostly sensitized to the change in power price and the discount rate. The investment appears to be financially viable under the 3 different scenarios, including a moderate growth, strong growth and stagnant growth rate of power prices. As the EU is moving aggressively towards a green economy, the two cases of moderate and strong growth are more realistic. In the upside case with a strongly optimistic view about electricity price, the net present value of the plant is approximately NOK 700 million. In the base case with moderate growth, the NPV is estimated to be roughly NOK 300 million.

As a sustainable investment consultant, my recommendation to your company is to proceed with the investment and pay particular attention to the environmental and social aspects of the project.

Introduction

This report is about a positive investment opportunity on Solar Park for the Norwegian venture capitalist firm. The firm is considering investing in solar panel systems through capital funding for a company specializing in ground-mounted PV power plants, with the belief that the project is going to generate economically and environmentally positive rate of return for the venture company. As the world is becoming more acknowledgeable about climate change, it is essential for more capital to flow into green sectors to ensure that the sustainability transition is on track. Norway is famous for its abundance of water resources, which makes the country a reliable place for consistent hydropower generation over the long term. However, electricity production from solar panels is relatively low due to the long cold Norwegian winter season with shorter daylight hours. This is one of the reasons why solar power today only produces one per thousand of the total power production in Norway. A recent report by the Energy Commission (2023) states that in order for Norway to reach the country's sustainability objectives in 2030, it needs around 55 million square meters of solar panels. This represents a huge investment opportunity for the venture capitalist firm who want to factor sustainable projects in their portfolio due to the increasing demand for solar energy in the country.

The venture capitalist firm will fund the company which specializes in ground-mounted PV power plants to construct, operate, and maintain a large-scale industrial power plant in exchange for an equity stake in the company. The facilities are used to generate electricity on a significant scale for industrial and commercial purposes, with the potential of providing energy for nearby communities. Total initial investment cost is estimated to be around NOK 500 million with the annual expected output of 130 million KWh, and the annual overhead cost of NOK 7 million. The project will be financed 80% through debt, with the borrowing rate at 6%, and the rest would be financed by the venture capitalist company. The lifetime of 30 years is also considered realistic by the company's suggestion. Moreover, the industrial power plant is located in the Western part of Norway, where sun coverage is similar to Germany, indicating that solar power generation will likely to be consistent and reliable. It is also worth noting that the plant is positively agreed by local politicians as they have promised to support the license application but receives strong opposition from the locals.

The report contains a discussion of advantages and disadvantages regarding the industrial power plant, a Due Diligence (DD) analysis of key sustainability considerations and a valuation of the project based on the Discounted Cash Flow valuation method. I will also discuss uncertainties related to the project and present a sensitivity and scenario analysis based on these uncertainties. Attached to this report is the excel model I have used to perform the valuation. Despite mentioning earlier that the plant will be used both industrially and commercially, the focus of my model is for commercial purposes. This takes into consideration that energy generated from the plant would be the power plant company's main revenue. For further requirements regarding the industrial use of the plant, such as power generating would save the alternative cost of energy purchasing from grids, I would be happy to adjust my models for better suitability of your company.

Advantages and Disadvantages of Solar Park

Solar panel systems are one of the common sources of renewable energy that provide clean and reliable energy and have the potential for greatly reducing greenhouse gas emissions. For this reason, solar energy is known for its climate change mitigation, which becomes very critical to protect humans, wildlife, and ecosystems. Despite its positive effect on reducing greenhouse gas emissions, solar energy can have negative impacts on the environment and local communities if the project has not been carefully planned and managed. Below, I present some key advantages and disadvantages of solar panel systems.

Key benefits include:

- *Clean and renewable energy production*. Solar energy uses sunlight to generate electricity, which reduces dependence on fossil fuels. They also produce no air pollution and emit no greenhouse gas during operation as well as require low water consumption.
- *Cost savings*. Once the plant is constructed, it costs nothing to produce energy, hence allowing for long-term saving cost on electricity purchased.
- *Positive return on investment*. Because solar energy saves the cost of purchasing power, it offers attractive financial return for investors.
- *Relatively long-life span*. Typically, 25 years or more.

Key disadvantages include:

- *High upfront cost.* Substantial development and installation costs.
- *Weather dependency.* Require sunlight to operate.
- *Significant land requirement.*
- *Environmental and ecological impacts.* Natural habitats, biodiversity of local environments is likely to be interrupted.
- *Limited energy generation at nighttime*

Opportunities and Risks

Opportunities

The investment in Solar Park offers many opportunities for the venture capitalist firm. The most important benefit is that the project aligns with Norway's sustainability goals towards the Energy Commission as solar energy offers significant environmental benefits. Solar power generates renewable energy from sunlight, which reduces the dependence of fossil fuels, minimizing air and water pollution. Therefore, it supports the reduction of greenhouse gas emission and mitigating climate change. By investing in solar panel system in Norway, your firm would not only contribute to Norway's goal of achieving carbon neutral society but also promoting sustainable investment activities for your business. This is also likely aligned with the EU's efforts to promote sustainable finance and its transition towards a greener economy.

Solar panel system is considered a strong fit within the EU's taxonomy sustainable investment framework, therefore investing in the project help demonstrate your company's commitment to complying with the sustainable finance regulations and staying ahead of evolving in the industry (European Commission, 2020). There is a positive impact of investing in projects that promote clean energy generation and climate change mitigation. The impact includes enhancing the venture capitalist firm's strong brand reputation and boosting its market competitiveness within the industry. Good brand image positions the company as a responsible and forward-thinking investing fund committed to addressing climate change.

There are several positive social impacts which solar panel systems can offer. The project does not only generate and ensure a sustainable source of energy, but it also creates employment opportunities to support the local economy and communities surrounding during the construction and maintenance phases. Moreover, communities living nearby may benefit from energy production through the reduction of their electricity bills in the forward-looking future. Being a responsible shareholder in the investment project means that the venture capitalist company should raise more voice and enhance dialogue with the solar panel company to ensure that the livelihoods of surrounding community is positively affected.

Investing in renewable energy in Norway offers many opportunities to receive strong government supports. Despite regulatory barriers and inconsistent policies over the past years has provided less incentives for solar panel system in Norway, I believe that Norway would have to expand their renewable energy production from solar energy, apart from hydropower, wind power and offshore wind, to be able to meet the country's Energy Commission target of 2030 (Regjeringen.no, 2023). Therefore, more various supporting programs including investment grants or tax incentives are more likely to be offered by the Norwegian government. The project will also be supported by local politicians in the area regarding license application. Over the long term, the project is promised to enhance the financial viability for the company.

Solar panel systems are also considered as a risk mitigation opportunity, especially if the company is seeking to diversify their portfolio in today's volatility. As the world is increasingly acknowledging the importance of climate mitigation, there are uncertainties regarding climate-change related risks such as regulatory risk, carbon pricing risk, and stranded assets which can impact the firm's traditional investment activities. By diversifying the firm's portfolio and investing in renewable energy projects, it can reduce the exposure to these risks and position itself in a long-term investment strategy.

Risks

Norway does receive an abundance of sunlight especially during summer months; however, the Western location of the plant indicates that it may experience frequent rainy and windy days with limited sunlight. These weather conditions are not favorable and can potentially lead to a decrease in energy production. Depending on the exact location of the PV park, strong winds can damage and disrupt the operation of solar panels. Therefore, mitigation measures regarding weather risks should be proposed by both parties, such as the incorporation of energy storage system to ensure a more reliable energy supply.

There is also an increasing recognition concerning the environmental and social impacts of such large-scale solar power plant, including land and hazardous materials from manufacturing processes (Tawalbeh et al., 2021). The PV Park requires a significant amount of land, specifically will cover 1000 daa which is approximately 170 soccer fields. The majority of the land is outfields and old logging fields that are currently being used for outdoor sheep grazing. Despite the field is at medium quality, the solar park may present an impact on local nature and habitats, as solar panels create shading on the ground beneath them and prevent sunlight from reaching vegetation. This shading can hinder growth of shade-tolerant plants and may potentially affect the high-quality grass-fed beef in the area. Hence, it is important to note that the venture capitalist firm should pay attention to the management plan of the solar power company. Considerations and management ought to be designed carefully to optimize the configuration and placement of solar panels to minimize shading and maximize sunlight exposure. One such adaptive measure is to integrate biodiversity enhancement through sheep grazing and creating pollinator-friendly habitats (Ramkumar, 2022). Sheep grazing has been proved to be great groundskeepers in the US when it comes to solar farms.

When examining across the life cycle and various categories of solar panel system, Hammerstad (2022) found that the production and construction stages represent the larger part of environmental impact. The manufacturing of solar cells contains various hazardous materials, and the production of these raw materials involves mining and several extracting processes. It is hence questionable regarding how these materials are disposed of and recycled at the end of their useful life. Even though there are several recycling processes that have been well established, these processes are energy intensive, quite complex and may utilize large quantity of chemicals (Tawalbeh et al., 2021). Since the company handling the Solar Park project specializes in ground-mounted PV power plants, it is my recommendation that the capitalist firm cooperates with the power plant company regarding waste disposal plan. Recycling PV waste and disposed PV modules is a crucial step to reduce the environmental impacts of solar power system.

Hammerstad (2022) also mentioned that 90-95% of all panel components are produced in China, and it raises a particular concern over the control of production and manufacturing stage. Long-distance supply chain results in unanswered questions related to the production aspect and management of supply chain. Particularly, human rights violation, forced labor and discrimination may pose a threat with respect to ESG screening and performance of the project. Norwegian solar power company owns a lot of products from China, and it is extremely important to report on ESG issues that are aligned with the EU's rules and regulations. The venture capitalist firm is also encouraged to communicate with the solar company regarding its supply chain management and considering other clean solar cells manufacturers to ensure clean and sustainable materials.

As mentioned earlier, local communities seem strongly opposed to the Solar Park project with the company. There is not only their concern over outdoor grazing but also the bad reputation and image of companies being involved in the project if the locals protest. These problems should be recognized and addressed through effective community engagement and transparent communication to gain local acceptance. Renewable projects usually require a need for effective reputation and stakeholder management, hence any negative incidents related to the ESG issues can harm the reputation of the project and the firms involved. It is extremely essential to implement robust ESG practices, effective engaging with stakeholders and maintaining transparency to mitigate reputational risks.

Valuation

Assumption

Discounted Cash Flow (DCF) model has been used for evaluating the Solar Park project. As a lot of assumptions go into the model, I'll start off by explaining how I deliver the discount rate (WACC) for the project.

The investment is financed with 80% debt of the initial cost, and the remaining 20% is financed through funding from the venture capitalist firm. This gives the debt-to-equity ratio of 4, relatively large since a large proportion of the investment is financed through debt. The average cost of debt is 6%, as provided by the company's information. The cost of debt is considered relatively high, especially for an investment in a developed economy like Norway. This suggests that the project may involve uncertainties throughout its lifetime, and that it can affect the performance of the investment.

I have used the Norwegian 10-year government bond which per this day is 3% as the risk-free rate based on the PwC report (2022). I also chose the market risk premium of 5,5% as of today from the same PwC report. An unleveraged industry beta of 0.64 is obtained from the website of professor Darmodaron for Green and Renewable energy sector in Western Europe. To reflect the company risk's profile and include its capital structure, the beta is relevered to 2.64. High beta is mainly driven by the high debt-to-equity ratio. The require rate of return is calculated using the CAPM model, which gives a rate of 18%. Based on all the assumptions above, WACC is estimated at **7.24%.** This rate represents a medium risk profile and is quite reasonable, especially when environmental and social issues discussed above are the main concerns for the Solar Park.

Apar from the discount rate, other assumptions are also included:

- An initial investment of NOK 500 million.
- An annual overhead cost accounts for the operation and maintenance of the plant, which is NOK 7 million. Depending on the location and the size of the plant, this cost may vary in the future.
- Project lifetime of 30 years is considered reasonable by the solar power company, as supported by different sources (NREL, n.d.).

- Linear depreciation is normally used in project valuation, which equals to NOK 16.67 million.
- Expected production output of 130 GWh, with 0.5% efficiency loss over its useful lifetime. The efficiency loss of solar panel system is supported by several sources (Chowdhury et al., 2020).
- Power price is estimated at 0.39kr/KWh, together with my initial assumption in the introduction that energy generated from the Solar Park will be used for commercial purposes. The price is calculated by taking the 5-year average spot price (2017-2021) of Bergen area from Nord Pool (Nord Pool, n.d.). Price for 2022 has been much more volatile due to the Russa-Ukraine war, and this may exaggerate the project's profitability due to the higher power price. Therefore, I have chosen the 2017-2021 period to account for some stabilities. I also add 2.5% increase in price growth for until 2030, and a cubic price growth from 2030 due to the strong belief that higher demand for renewable energy would drive the prices up. CO_2 price is also factored in the electricity price within the EU area, hence, the transition towards a green economy would increase CO_2 price as well as power price. My suggestion regarding the growth rate of electricity price is also supported by NVE power price forecast report (2017).
- Norwegian corporate tax rate 22%.

It should also be mentioned that electricity generated from the Solar Park can be utilized directly by industrial entities nearby. These may include manufacturing facilities, office buildings, shopping centers, and any other energy-intensive operations. One could have included it as the cost of saving electricity if purchased from grid. However, I have chosen not to do so since this report would focus on commercial use only.

Discounted Cash Flow (DCF)

The discounted cash flow analysis suggests that the project obtains a NPV value of NOK 322 million with the Internal Rate of Return (IRR) of 11.22%. Figure 1 presents the net yearly cashflow and projected revenue from the investment over its useful lifetime. The graph shows that the Solar Park project is projected to generate sufficient cashflow to cover its expenses since year 11th, and that expected revenue from that year exceeds NOK 100 million.

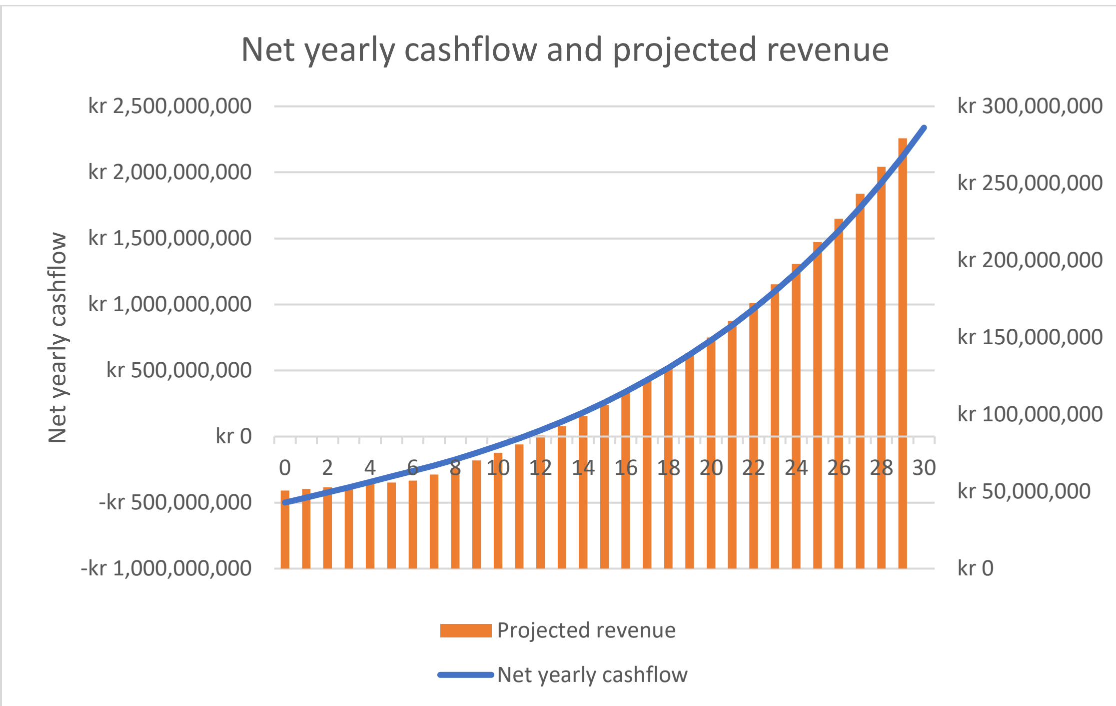

Figure 1. Net yearly cashflow and projected revenue

As a lot of assumptions go into the DCF model, the valuation is likely to be affected by several uncertainties. It is thus important to address these uncertainties regarding how our model would be subjected to change. This is where sensitivity and scenario analysis come in, whose sensitivity analysis allows for forecasting uncertainties by studying all the possible outcomes of when adjusting some important variables in the model.

Some of the key uncertainties I believe related to:

- Uncertainty about the discount rate due to the medium-risk profile of the project regarding environmental and social issues.
- Uncertainty about the expected power price growth due to higher demand for renewable energy.
- Uncertainty for the expected production output regarding weather risk
- Uncertainty of the initial investment cost and production cost because of technological advancement and waste disposal cost.

Sensitivity Analysis

A. Discount rate

The discount rate contains many assumptions in the calculation, it also represents the project's risk profile or the cost of capital of the investment. The discount rate incorporates not only the perceived risk of the investment but can also be influenced by broader market conditions. As mentioned earlier, despite solar system emitting no greenhouse gas during its operation, the project has some underlying risks regarding the environmental impact on vegetation and natural habitat. Strongly local opposition against the Solar Park can also have an impact on the long-term profitability of company.

The initial discount rate being used in the model is 7.24%, this rate is relatively high, mainly driven by the large debt-to-equity ratio of the project. The loan rate for which the investment has is 6%, this rate is again quite high considering a well-developed economy like Norway. The high loan rate is possibly reflected by the uncertain economic outlook regarding high interest rate of post-Covid period and the Russia-Ukraine war effect. Market turmoil threatening to undermine the efforts to curb inflation in several countries, followed by the collapse of a few US banks and the depreciation of the Norwegian currency has signaled for consistent market uncertainties in the next few years.

Responding to these potential risks, I adjusted some changes in the discount rate to examine its impact on the value of NPV. With a change of +2% in the discount rate, for example driven by the strong local opposition and continuous high interest rate, NPV value from Figure 2 drops to more than 40% compared to its initial value of NOK 322 million. It is important to note that NPV remains positive corresponding to the change in discount rate.

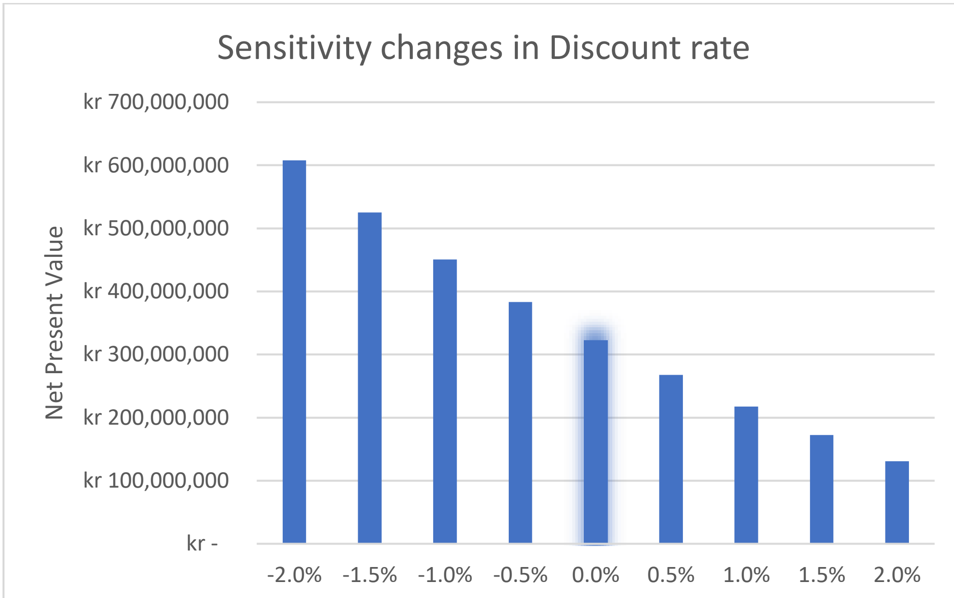

Figure 2. Discount rate sensitivity.

B. Expected power price growth

In the initial assumption, power price is expected to growth by 2.5% every year until 2030 and cubic growth afterwards, as projected by NVE future price forecast (2017). The renewable energy sector is growing much faster than it has ever been in Europe, and the growing demand for the sector will likely drive electricity prices. Norway has been utilizing most of its resources to produced renewable energy, and the government's commitment to electrification transition would be the main reason for higher power price in the following years.

Figure 3 shows the changes in the net present value with the changes in power price growth. As we can see, it becomes more forward-looking for the future financial viability of the project as power price grows consistently. Even though a fall of 2% in the price growth will be less likely in near future, it is worth noting that NPV would have a negative value.

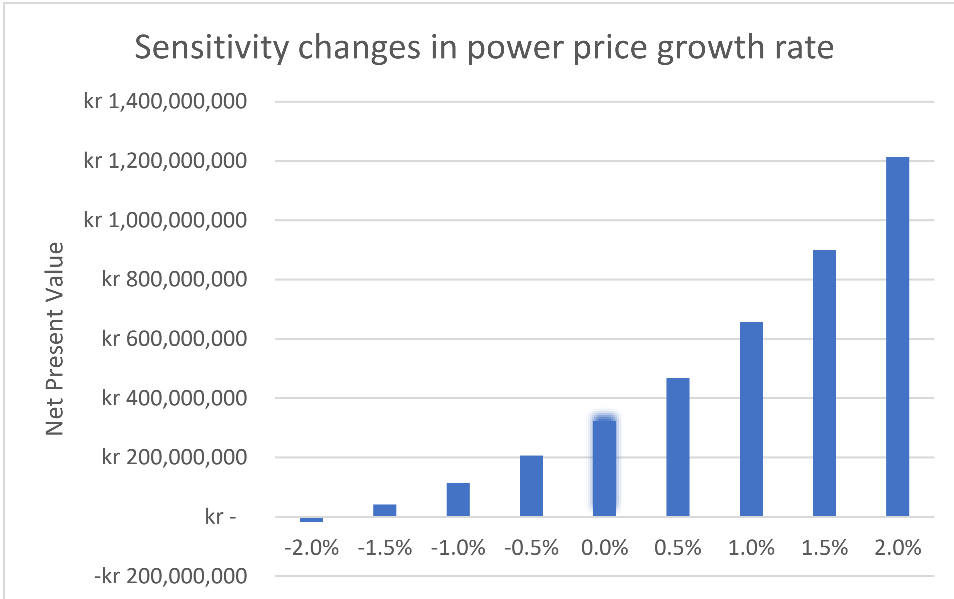

Figure 3. Power price growth rate sensitivity.

C. Expected production output

As mentioned, several times in the report, Norway has it advantage for long daylight hours during summer, which make its more profitable to operate the Solar Park. However, the country is also known for its dark and cold winter season, and the fact that the project is in the Western part presents uncertainty regarding weather conditions. There may be more output expected in the summer and less production during winter, resulting in inconsistent financial capability of the investment. For this reason, I added some changes in the production output to see how large these impacts can have on the NPV value. Higher output clearly increases the financial viability of the Solar Park, while a fall of 40% due to less sunlight or efficiency loss during the lifetime leads to negative NPV.

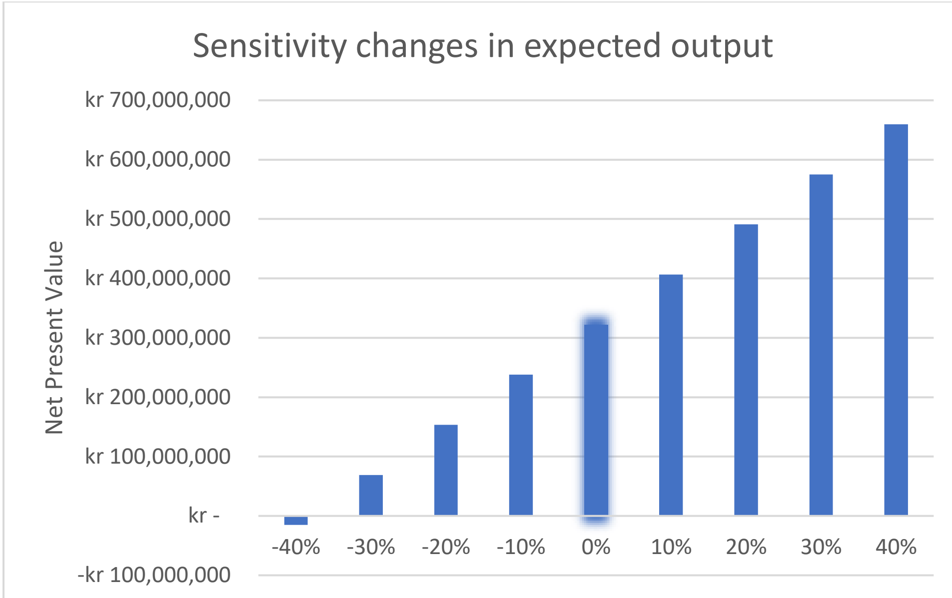

Figure 4. Expected output growth rate sensitivity.

D. Initial investment

Despite solar PV installation cost has declined steadily over years, it is considered relatively high in Norway. Most of the solar cells in many Norwegian solar power companies are imported from China, which are not only about the concerns over ESG issues in the manufacturing phase but also the geographical distance making it more costly (Harmmerstad, 2022). There is also a concern regarding the waste management process of the materials at the end of their useful life. These problems could pose a threat to the increasing initial investment cost, e.g. requiring more R&D. Weather risks such as strong wind and storm could also disrupt the Solar Park facilities, therefore requiring materials replacement. As figure 5 suggests, an increase of 20% in the initial investment cost results in a fall of 70% of the NPV. Over the course of the long-term when more R&D process has been done and the technology matures, I expect the NPV of the project to be relatively larger.

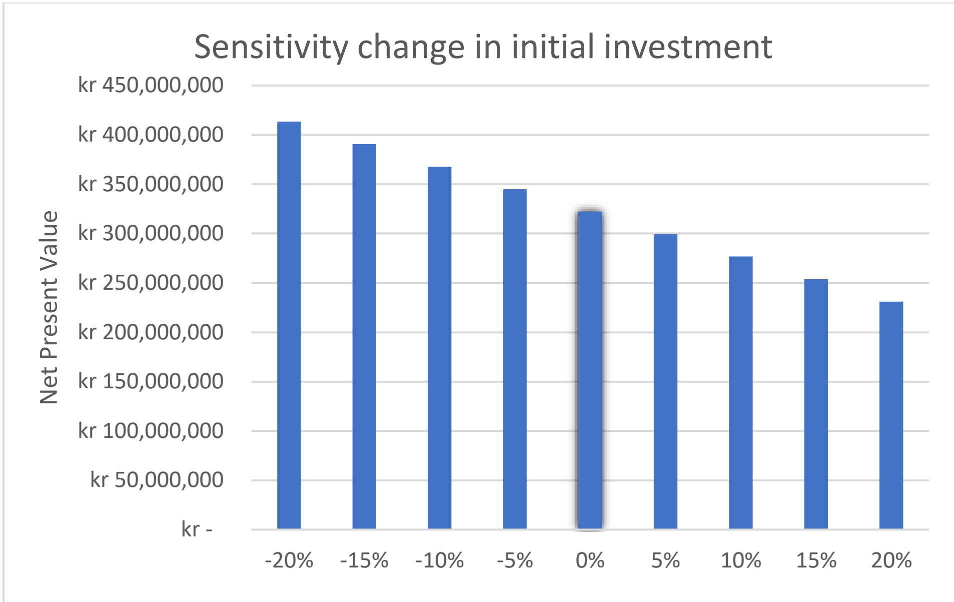

Figure 5. Initial investment sensitivity.

E. Overview

In a broader picture, an overview of how large these impacts have influenced the changes in NPV is presented below in Figure 6. The graph indicates that power price growth is the most sensitive variable that changes the NPV. A change of -1/+1% in the expected price growth could make the NPV move up and down by almost 65%. The second most sensitive variable is followed by the discount rate, whose -1/+1% change results in a 40% change in NPV. The expected production output and the initial investment have less impact on the change of NPV, which are 13% and 7% respectively.

Finally, the total production cost has also been assessed, potentially due to higher maintenance costs resulting from faster degradation of the plant because of weather disruption. However, the project seems to be less sensitive to this change.

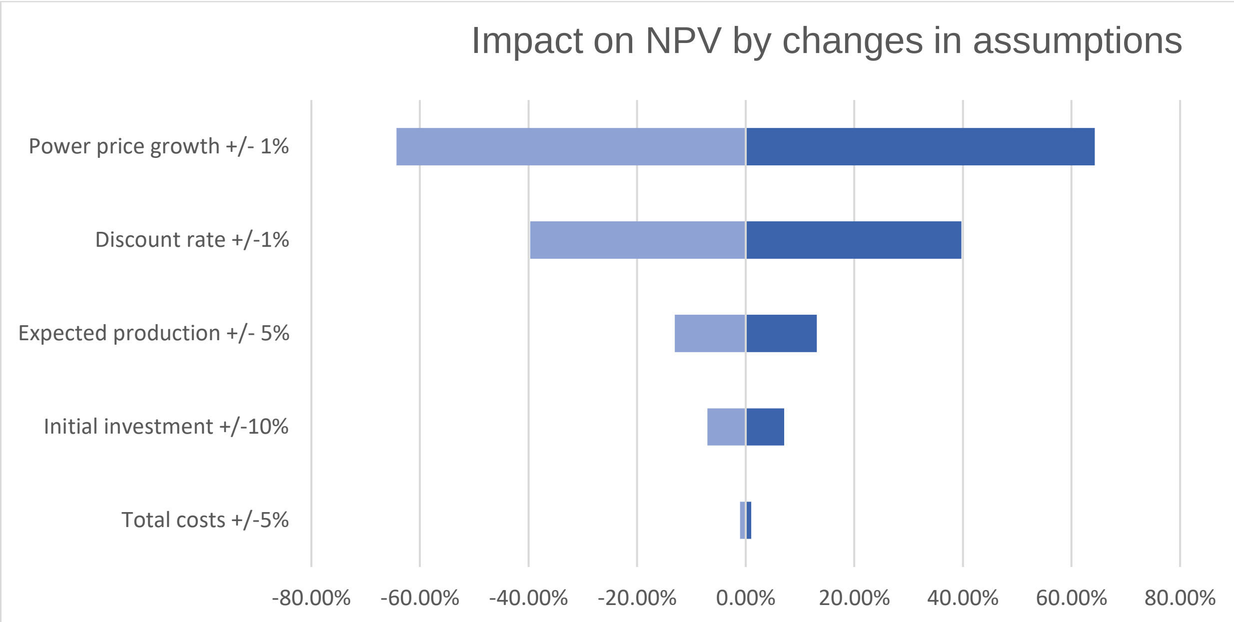

Figure 6. Impact on NPV by changes in assumptions

Scenario Analysis

As previously discussed, the financial model contains different assumptions, and several uncertainties are associated with these assumptions. In this part, I present different scenarios related to the Solar Park project, including an upside case and a downside case. The outcome of these scenarios is presented under table 1.

Table 1. Net present value and the Internal Rate of Return in different scenarios

	Base case	Upside case	Downside case
NPV	322,170,827 kr	707,821,068 kr	4,388,948 kr
IRR	11.22%	14.05%	7.32%

In the upside case, there is a strong optimistic view regarding the future growth rate of the power price, which resulting in a 2.5% increase in the price until 2030 and a quadratic growth rate afterwards. The upside case is supported by a strong belief that CO_2 will be priced higher in the future due to aggressive plan and execution of the EU towards the green economy. Higher renewable consumption regarding the higher demand of electrification in Norway also drives the power price higher. Together with the aggressive growth rate of power price, output from the Solar Park is also expected to rise 15% in this case, assuming that technological advancements make the production and storage phase more efficient. With these assumptions in mind, NPV is estimated to be around NOK 707 million with a 14.05% IRR in the upside case.

There are several changes in the assumptions regarding the downside case including:

- Stagnant power price growth, with 2.5% increase annually over 30 years.
- 10% reduction in expected output.
- 10% rise in total cost.

The idea behind the reduction in expected output and the rise in total cost is mainly drawn from uncertainties regarding weather fluctuation, resulting in more maintenance and monitoring activities for the aging facilities. The Norwegian company that constructs the Solar Park specializes in ground-mounted PV, however, Norway today only produces 0.225TWh solar energy. This indicates that the industry is not yet matured to ensure stability in the country. Therefore, uncertainties about the increase in maintenance cost and the fall in expected production are to be expected. The downside case results in an NPV of NOK 4 million and an IRR of 7.32%.

Conclusion and Recommendation

The valuation results in a positive net present value and a positive internal rate of return, and it is my recommendation as a consultant that your company should proceed with Solar Park. Despite the project represents some ESG issues related to vegetation and strong local opposition, solar panel system offers various opportunities including no green house gas emission during its operation, and high alignment with Norway's national sustainability goals and the EU taxonomy. As outlined in the report, the power price is the most sensitized variable in the model, suggesting that the project is more likely to be profitable in the future as the demand for renewable energy rises. However, it is extremely important for your company to pay careful attention to the engagement with the locals regarding the environmental impact of the project. With careful planning and execution, both the venture capitalist firm and the solar power company can help prevent unwanted bad reputation.

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